

БЕЗОПАСНОСТЬ ОПЕРАЦИОННЫХ СИСТЕМ

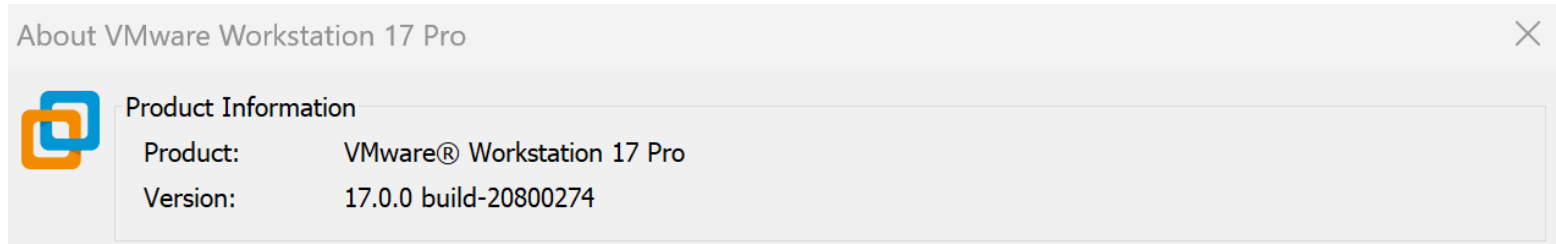
ПРАКТИЧЕСКОЕ ЗАДАНИЕ: Настройка Windows Server 2019

Выполнил: Юрий Шамрай (MIFIB/1-й поток)

1. Установите Windows Server 2019

В качестве гипервизора, для создания виртуальной машины с **Windows Server 2019**,

используем **VMware Workstation 17 Pro**:



со следующими параметрами:

OS: Windows Server 2019 Datacenter Evaluation (Desktop Experience)

CPU: 1CPU x 4vCORES

RAM: 8 GB

HDD: 60GB

LAN: Gigabit (10/100/1000baseT) Ethernet adapter.

CD/DVD: для установки со скаченного ISO-образа

DISPLAY: Максимальное разрешение (опционально)

Device specifications

Device name	W2K19-PROSPERO
Processor	11th Gen Intel(R) Core(TM) i9-11980HK @ 2.60GHz 3.30 GHz
Installed RAM	8.00 GB
Device ID	BCFB2AFE-3F7D-4274-9C9B-149090E86914
Product ID	00431-20000-00000-AA143
System type	64-bit operating system, x64-based processor
Pen and touch	No pen or touch input is available for this display

Rename this PC

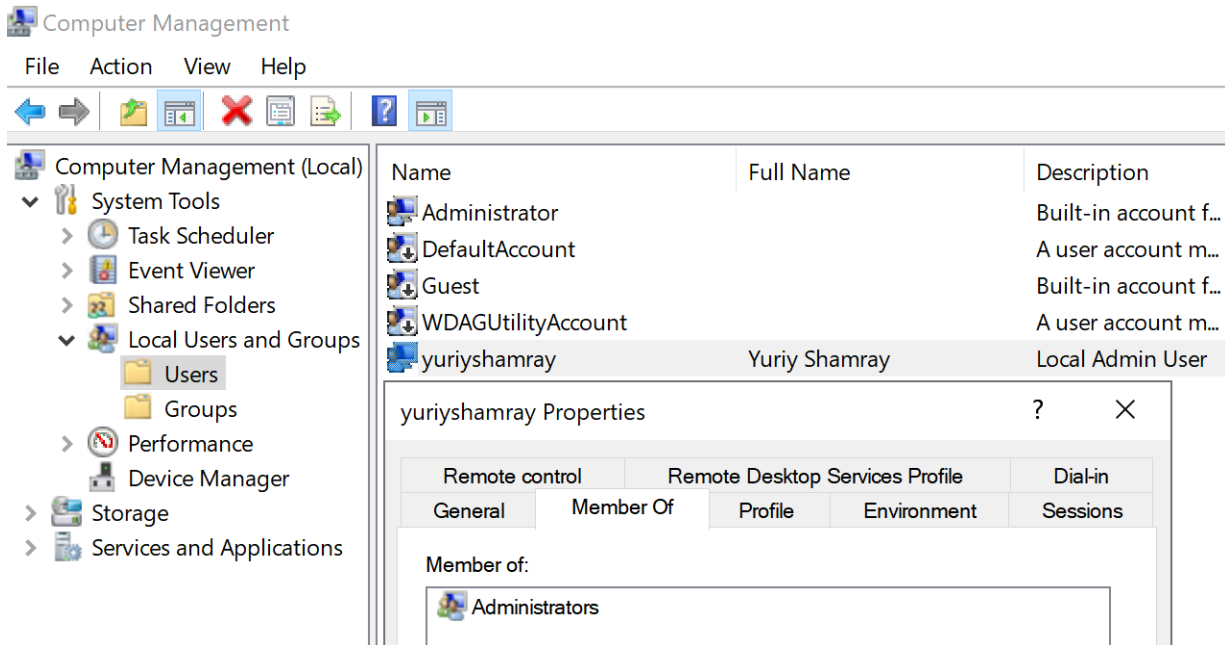
Windows specifications

Edition	Windows Server 2019 Datacenter Evaluation
Version	1809
Installed on	28 Mar 2023
OS build	17763.4131

Pic#1. Guest System => About

2. Создайте учётную запись с правами администратора. Укажите ваше имя и фамилию слитно латинскими буквами. Сделайте скриншот.

#GUI (Pic#2):



Pic#2. Computer Management => User Properties

#POWERSHELL (Pic3):

Все действия выполняем в Windows PowerShell (с правами Администратора)

Присвоим переменную **\$Password** для пароля и преобразуем в защищенную строку:

```
$Password = Read-Host -AsSecureString
```

вводим пароль для создаваемой учетной записи (скрыто *):

```
2023@Mifiib
```

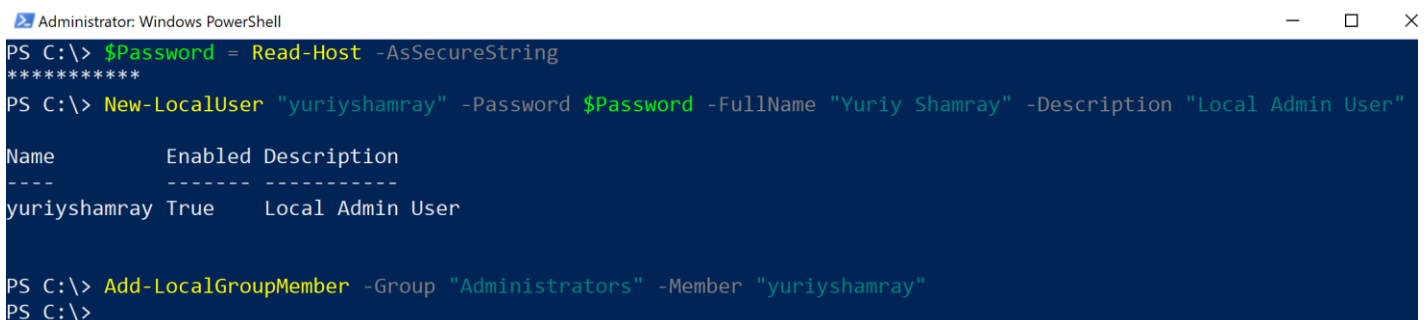
Создаём пользователя с именем **"yuriyshamray"**

Подставляем переменную **\$Password** в качестве пароля для этого пользователя:

```
New-LocalUser "yuriyshamray" -Password $Password -FullName "Yuriy Shamray" -Description "Local Admin User"
```

Добавляем его в локальную группу **"Administrators"**:

```
Add-LocalGroupMember -Group "Administrators" -Member "yuriyshamray"
```

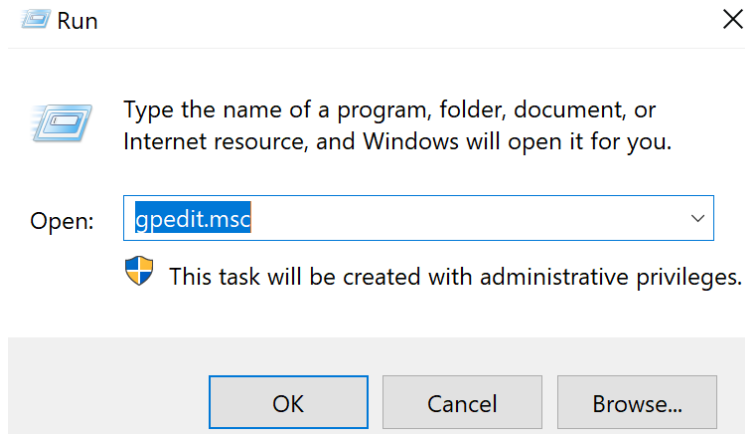


Pic#3. New-LocalUser; Add-LocalGroupMember

3. Настройте автообновление версий и патчей сервера.

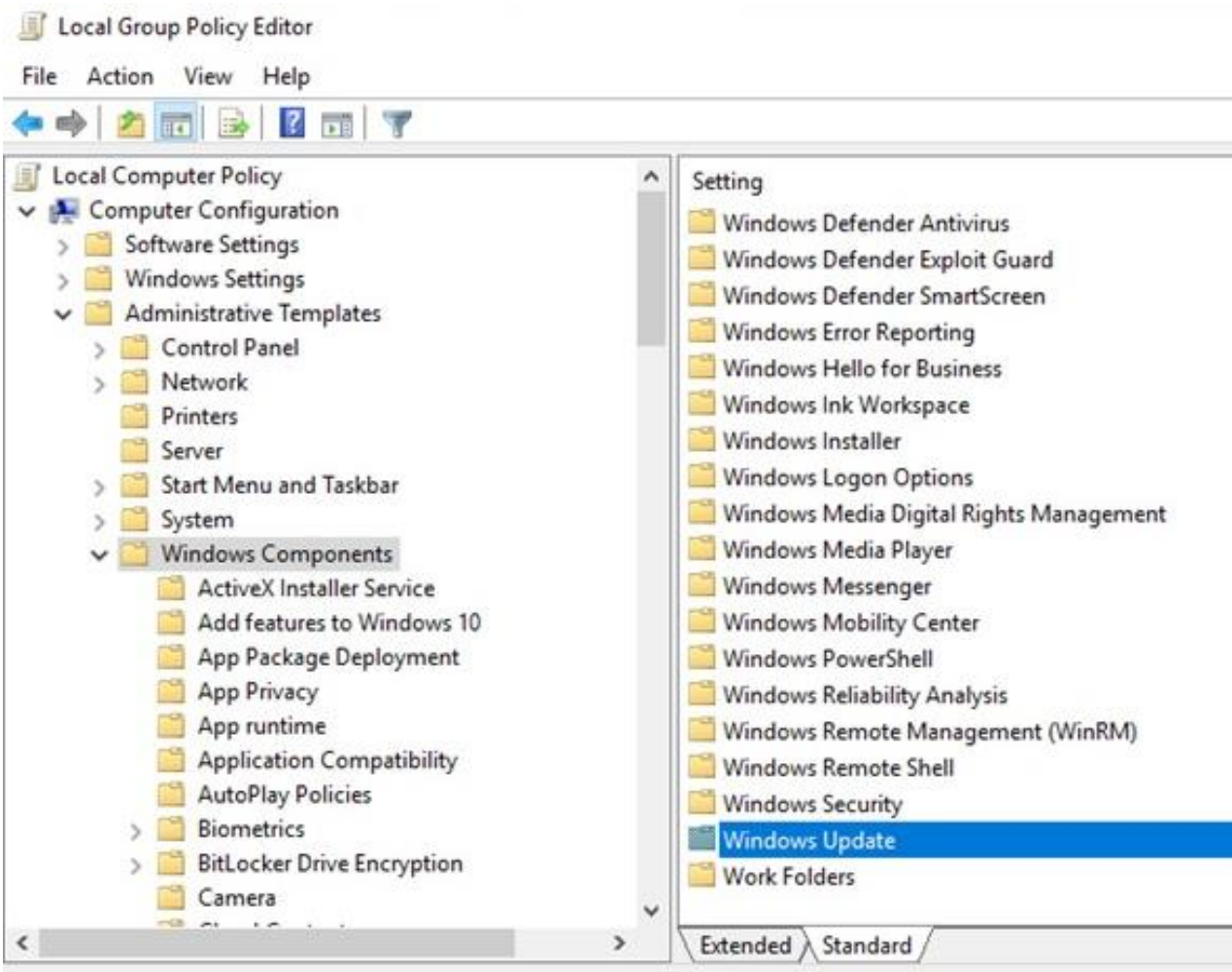
#GUI Group Policy Editor:

В окне **Run** вводим **gpedit.msc** (Pic#4):



Pic#4. Start => Run

Далее, в окне **Local Group Policy Editor**, переходим к следующим параметрам (Pic5):
Local Computer Policy => Computer Configuration => Administrative Templates => Windows Components => Windows Update



Pic#5. Local Group Policy Editor

В разделе **Windows Update**, включим и настроим следующие политики (Pic#6):

Setting	State	Com...
Windows Update for Business		
Do not display 'Install Updates and Shut Down' option in Shut Down Windows ...	Not configured	No
Do not adjust default option to 'Install Updates and Shut Down' in Shut Down ...	Not configured	No
Enabling Windows Update Power Management to automatically wake up the s...	Not configured	No
Turn off auto-restart for updates during active hours	Not configured	No
Specify active hours range for auto-restarts	Not configured	No
Allow updates to be downloaded automatically over metered connections	Not configured	No
Always automatically restart at the scheduled time	Not configured	No
Specify deadline before auto-restart for update installation	Not configured	No
Configure auto-restart reminder notifications for updates	Not configured	No
Turn off auto-restart notifications for update installations	Not configured	No
Configure auto-restart required notification for updates	Not configured	No
Configure Automatic Updates	Enabled	No
Specify deadlines for automatic updates and restarts	Not configured	No
Specify intranet Microsoft update service location	Not configured	No
Automatic Updates detection frequency	Enabled	No
Do not allow update deferral policies to cause scans against Windows Update	Not configured	No
Remove access to "Pause updates" feature	Not configured	No
Remove access to use all Windows Update features	Not configured	No
Do not connect to any Windows Update Internet locations	Not configured	No
Allow non-administrators to receive update notifications	Not configured	No
Specify Engaged restart transition and notification schedule for updates	Not configured	No
Do not include drivers with Windows Updates	Not configured	No
Turn on Software Notifications	Not configured	No
Allow Automatic Updates immediate installation	Enabled	No
Turn on recommended updates via Automatic Updates	Enabled	No
No auto-restart with logged on users for scheduled automatic updates installa...	Not configured	No
Re-prompt for restart with scheduled installations	Not configured	No
Delay Restart for scheduled installations	Not configured	No
Reschedule Automatic Updates scheduled installations	Not configured	No
Configure auto-restart warning notifications schedule for updates	Not configured	No
Update Power Policy for Cart Restarts	Not configured	No
Enable client-side targeting	Not configured	No
Allow signed updates from an intranet Microsoft update service location	Not configured	No
Display options for update notifications	Not configured	No

Pic#6. Windows Update (Turn on recommended updates via Automatic Updates – опционально)

В политике «**Configure Automatic Updates**», выбираем вариант «**4 – Auto download and schedule the install**», установим флажок «**Install during automatic maintenance**» (также, мы можем указать дни и время установки (Pic#7):

Configure Automatic Updates

Previous Setting

Next Setting

Not Configured

Enabled

Disabled

Comment:

Supported on:

Windows XP Professional Service Pack 1 or At Least Windows 2000 Service Pack 3

Options:

Help:

Configure automatic updating: 4 - Auto download and schedule the install

The following settings are only required and applicable if 4 is selected.

Install during automatic maintenance

Scheduled install day: 0 - Every day

Scheduled install time: Automatic

If you have selected "4 - Auto download and schedule the install" for your scheduled install day and specified a schedule, you also have the option to limit updating to a weekly, bi-weekly or monthly occurrence, using the options below:

Every week

First week of the month

Second week of the month

Third week of the month

Fourth week of the month

Install updates for other Microsoft products

Specifies whether this computer will receive security updates and other important downloads through the Windows automatic updating service.

Note: This policy does not apply to Windows RT.

This setting lets you specify whether automatic updates are enabled on this computer. If the service is enabled, you must select one of the four options in the Group Policy Setting:

2 = Notify before downloading and installing any updates.

3 = (Default setting) Download the updates automatically and notify when they are ready to be installed

Windows finds updates that apply to the computer and downloads them in the background (the user is not notified or interrupted during this process). When the downloads are complete, users will be notified that they are ready to install. After going to Windows Update, users can download and install any available updates.

4 = Automatically download updates and install them on the schedule specified below.

When "Automatic" is selected as the scheduled install time, Windows will automatically check, download, and install updates. The device will reboot as per Windows default settings unless configured by group policy. (Applies to Windows 10, version 1809 and higher)

Specify the schedule using the options in the Group Policy Setting. For version 1709 and above, there is an additional choice of limiting updating to a weekly, bi-weekly, or monthly occurrence. If no schedule is specified, the default schedule for all installations will be every day at 3:00 AM. If any updates require a restart to complete the installation, Windows will restart the computer automatically. (If a user is signed in to the computer when Windows is ready to restart, the user will be notified and given the option to delay the restart.)

On Windows 8 and later, you can set updates to install during automatic maintenance instead of a specific schedule. Automatic maintenance will install updates when the computer is not in use and avoid doing so when the computer is running on battery power. If automatic maintenance is unable to install updates for 2 days, Windows Update will install updates right away. Users will then be notified about an upcoming restart, and that restart will only take place if there is no potential for accidental data loss.

5 = Allow local administrators to select the configuration mode that Automatic Updates should notify and install updates. (This option has not been carried over to any Win 10 Versions)

With this option, local administrators will be allowed to use the Windows Update control panel to select a configuration option of their choice. Local administrators will not be allowed to disable the configuration for Automatic Updates.

If the status for this policy is set to Disabled, any updates that are available on Windows Update must be downloaded and installed manually. To do this, search for Windows Update using Start.

If the status is set to Not Configured, use of Automatic Updates is not specified at the Group Policy level. However, an administrator can still configure Automatic Updates through Control Panel.

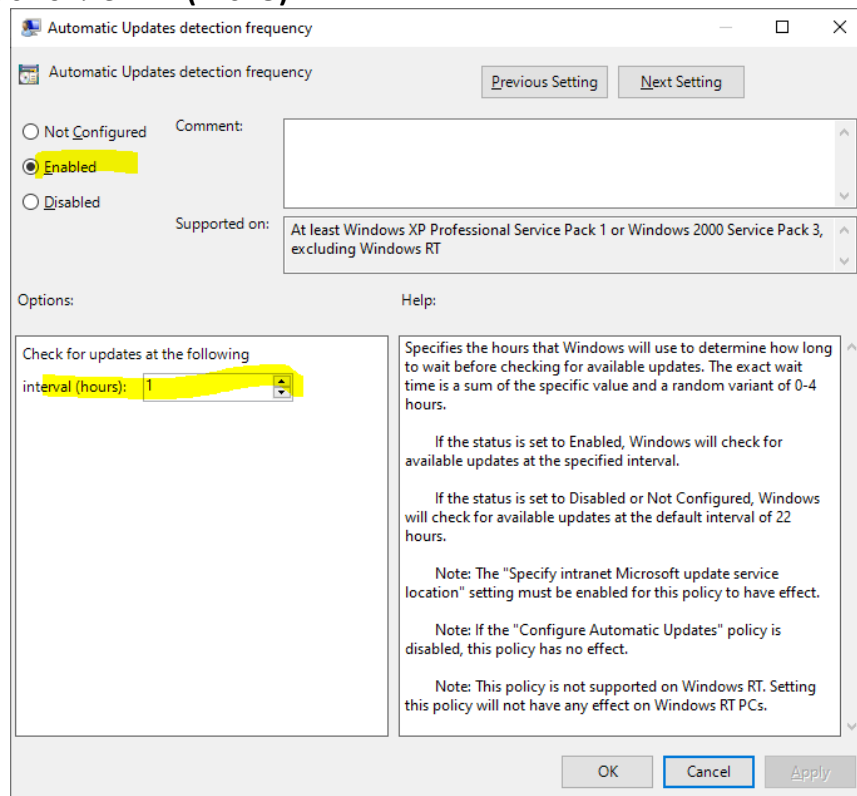
OK

Cancel

Apply

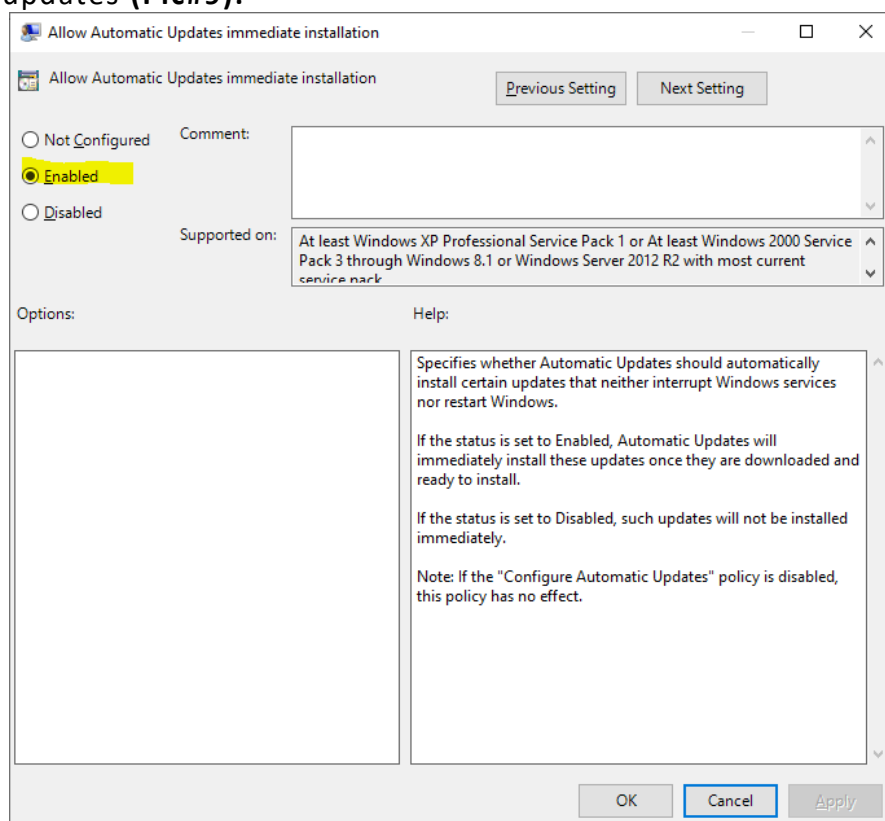
Pic#7. Configure Automatic Updates policy

Включаем и изменяем политику - **Automatic Updates Detection Policy** (Политика автоматического обнаружения обновлений). Эта политика определяет, как часто ОС будет проверять наличие обновлений (**Pic#8**):



Pic#8. Automatic Updates Detection Policy

Включаем политику «**Allow Automatic Updates immediate installation**» (Разрешить автоматическому обновлению немедленную установку). Это позволит устанавливать обновления, которые не прерывают работу служб Windows и не требуют перезагрузки, например, Windows Defender Definition updates (**Pic#9**):



Pic#9. Allow Automatic Updates immediate installation

Сделайте скриншот окна обновлений (Pic#10).

Windows Update

*Some settings are managed by your organization

[View configured update policies](#)



You're up to date

Last checked: Today, 8:34 PM

[Change active hours](#)

[View update history](#)

[Advanced options](#)

Policies set on your device

Get updates for other Microsoft products

Source: Administrator

Type: Group Policy

Set Automatic Update options

Source: Administrator

Type: Group Policy

Specifies the hours before checking for updates

Source: Administrator

Type: Group Policy

Pic#10. Windows Update

#POWERSHELL Core Mode (Pic11):

В командной строке вводим **sconfig** (конфигурация сервера), чтобы запустить утилиту настройки сервера Windows (**Pic#11**):

Administrator: Windows PowerShell

Microsoft (R) Windows Script Host Version 5.812

Copyright (C) Microsoft Corporation. All rights reserved.

Inspecting system...

Server Configuration

- | | |
|---------------------------------|----------------------|
| 1) Domain/Workgroup: | Workgroup: WORKGROUP |
| 2) Computer Name: | W2K19-PROSPERO |
| 3) Add Local Administrator | |
| 4) Configure Remote Management | Enabled |
| 5) Windows Update Settings: | DownloadOnly |
| 6) Download and Install Updates | |
| 7) Remote Desktop: | Disabled |
| 8) Network Settings | |
| 9) Date and Time | |
| 10) Telemetry settings | Unknown |
| 11) Windows Activation | |
| 12) Log Off User | |
| 13) Restart Server | |
| 14) Shut Down Server | |
| 15) Exit to Command Line | |

Enter number to select an option:

Pic#11. Sconfig => Server Configuration

Чтобы изменить конфигурацию Центра обновления Windows, вводим 5 **(Pic#12)**:

```
Administrator: Windows PowerShell
Microsoft (R) Windows Script Host Version 5.812
Copyright (C) Microsoft Corporation. All rights reserved.

Inspecting system...

=====
                        Server Configuration
=====

1) Domain/Workgroup:                Workgroup:  WORKGROUP
2) Computer Name:                   W2K19-PROSPERO
3) Add Local Administrator
4) Configure Remote Management      Enabled

5) Windows Update Settings:         DownloadOnly
6) Download and Install Updates
7) Remote Desktop:                  Disabled

8) Network Settings
9) Date and Time
10) Telemetry settings               Unknown
11) Windows Activation

12) Log Off User
13) Restart Server
14) Shut Down Server
15) Exit to Command Line

Enter number to select an option: 5

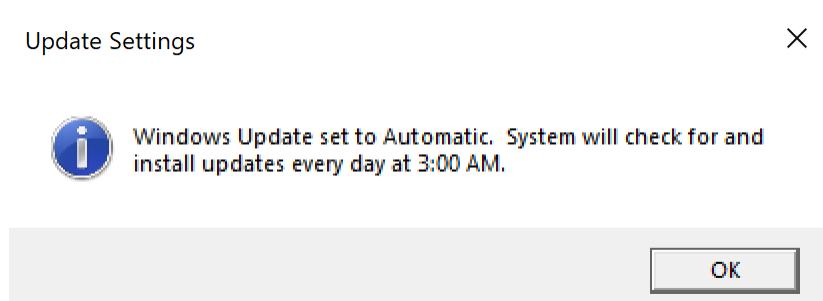
Windows Update currently set to: DownloadOnly
Select (A)utomatic, (D)ownloadOnly or (M)anual updates: a
```

(Pic#12). Access the Windows Update configuration.

Далее, вводим букву **а** для выбора режима **(A)utomatic**,

Появится окно о том, что режим обновлений Windows будет работать в автоматическом режиме.

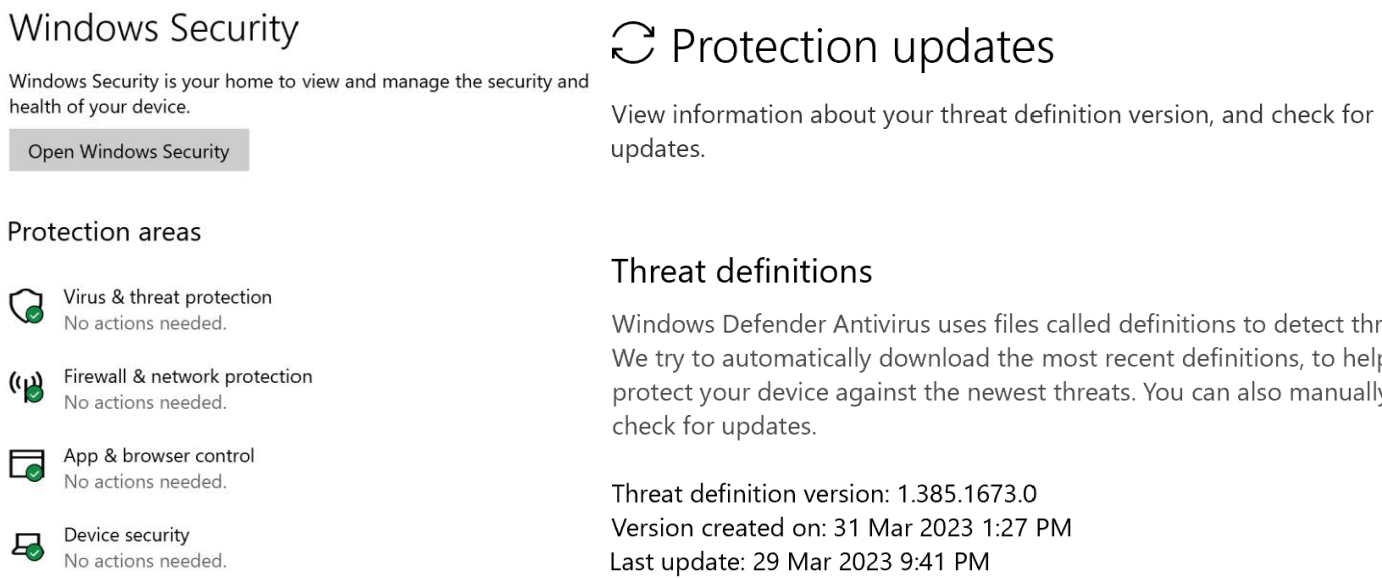
Система будет ежедневно проверять наличие новых обновлений и, если они есть, будет автоматически их устанавливать по расписанию в 3:00 AM системного времени **(Pic#13)**:



(Pic#13). Update Settings

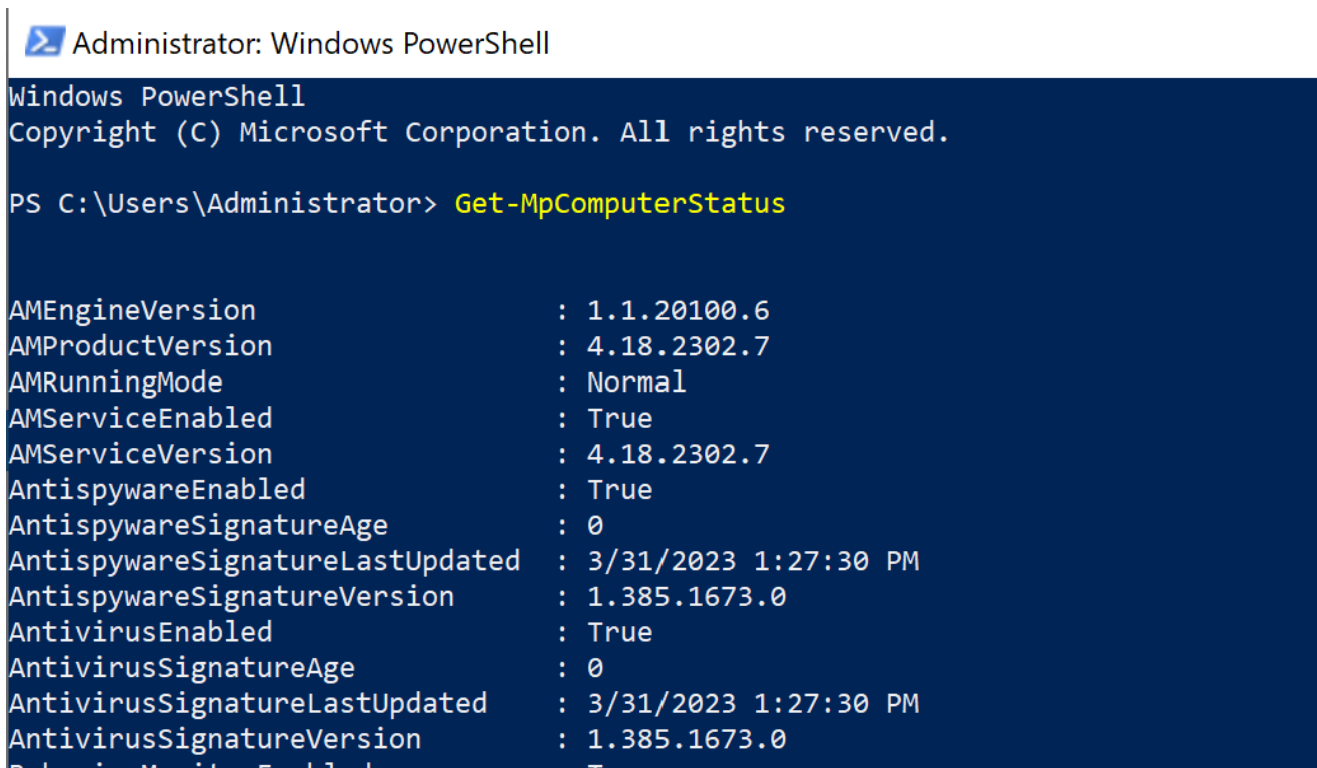
4. Проверьте, активен ли встроенный антивирус и установлены ли все актуальные обновления (любым способом). Сделайте скриншот.

GUI: Переходим **Start=>Settings=>Windows Security**. Проверяем, что встроенный антивирус активен, установлены актуальные обновления и нет никаких предупреждений (**Pic#14**):



(Pic#14). Windows Security GUI

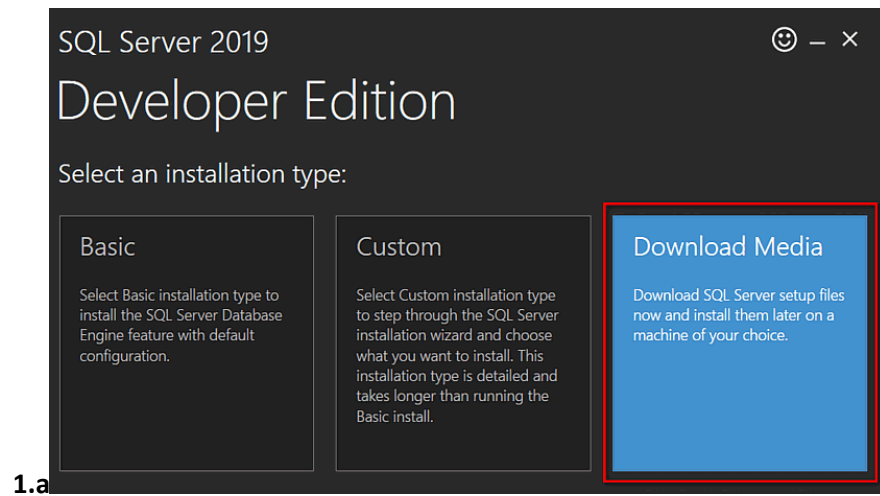
POWERSHELL: вводим команду **Get-MpComputerStatus**, строка **AMRunningMode - Normal** означает, что антивирусная программа Microsoft Defender работает в активном режиме. (**Pic#15**):



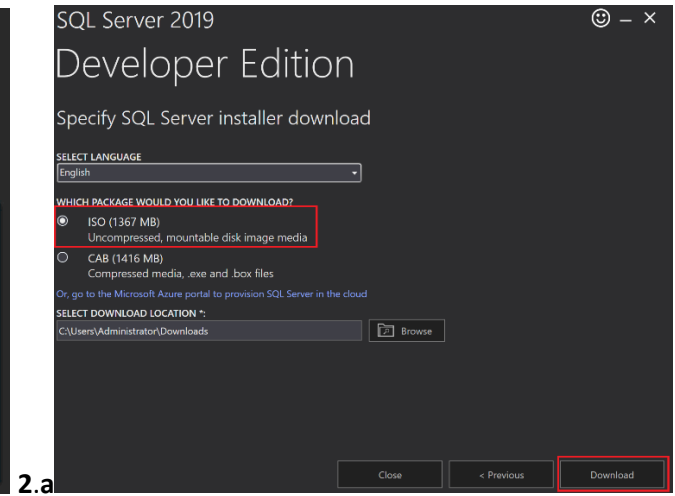
(Pic#15): Get-MpComputerStatus

5. Установите SQL Server 2019. Запустите программу по созданию записей в базе и подтвердите, что всё готово. Сделайте скриншот.

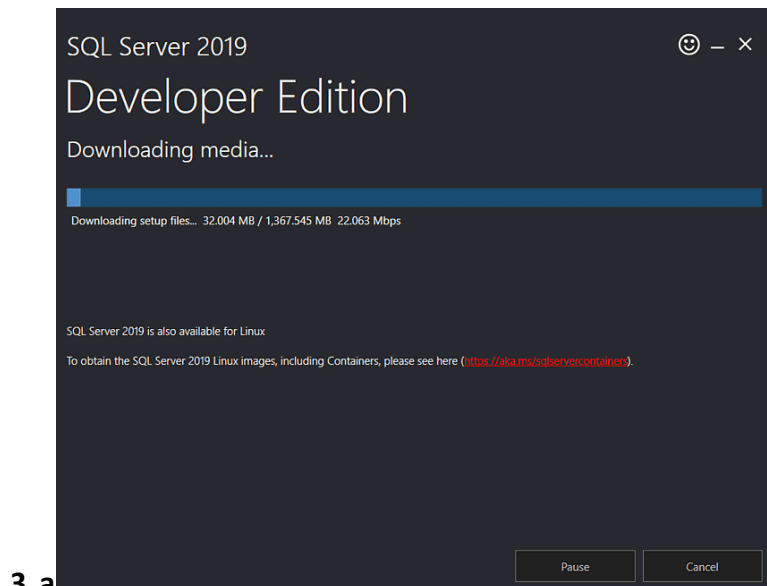
Установим SQL Server 2019 Developer Edition, по ссылке скачиваем установщик =>
Download SQL Server 2019 Developer Edition запускаем, далее скриншоты с установкой (Pic#1a-16a) :



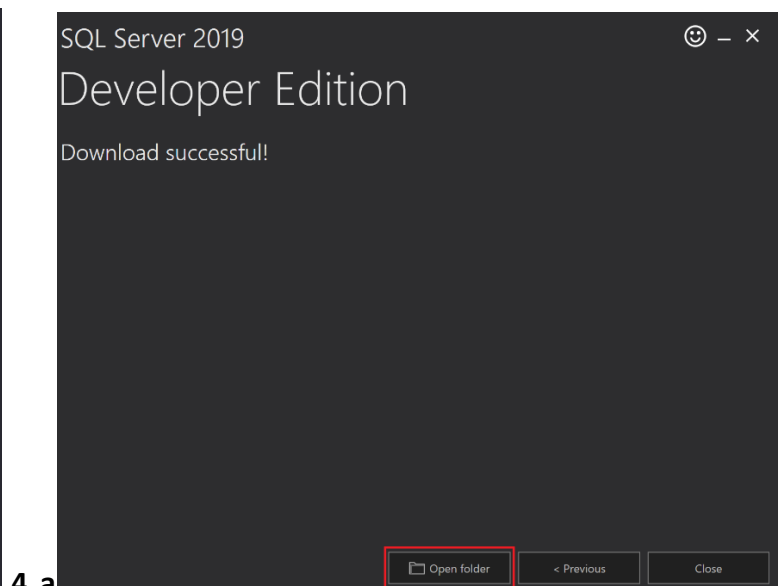
1.a



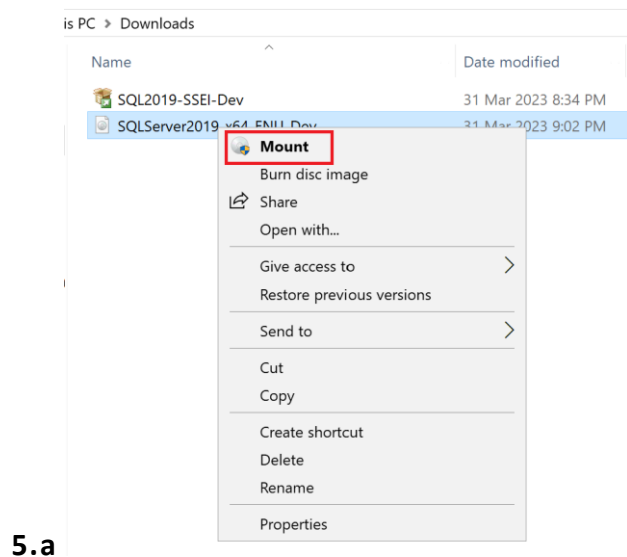
2.a



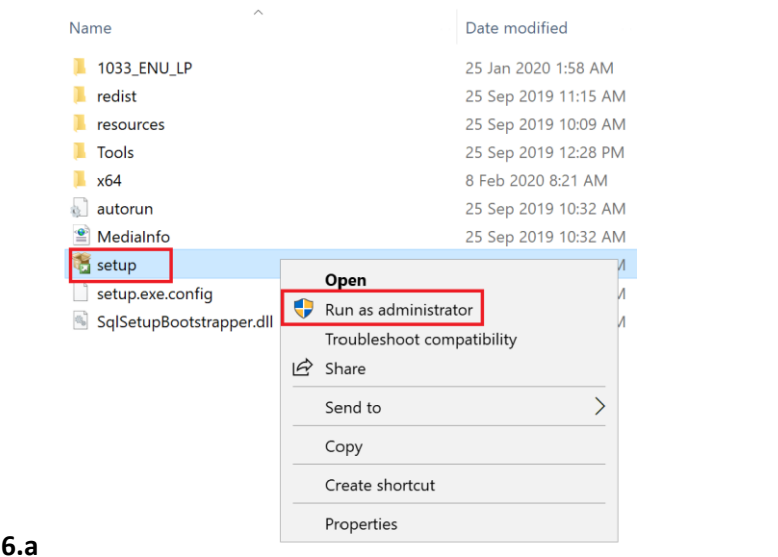
3.a



4.a



5.a



6.a

Planning

Installation

Maintenance

Tools

Resources

Advanced

Options

**New SQL Server stand-alone installation or add features to an existing installation**

Launch a wizard to install SQL Server 2019 in a non-clustered environment or to add features to an existing SQL Server 2019 instance.

**Install SQL Server Reporting Services**

Launch a download page that provides a link to install SQL Server Reporting Services. An internet connection is required to install SSRS.

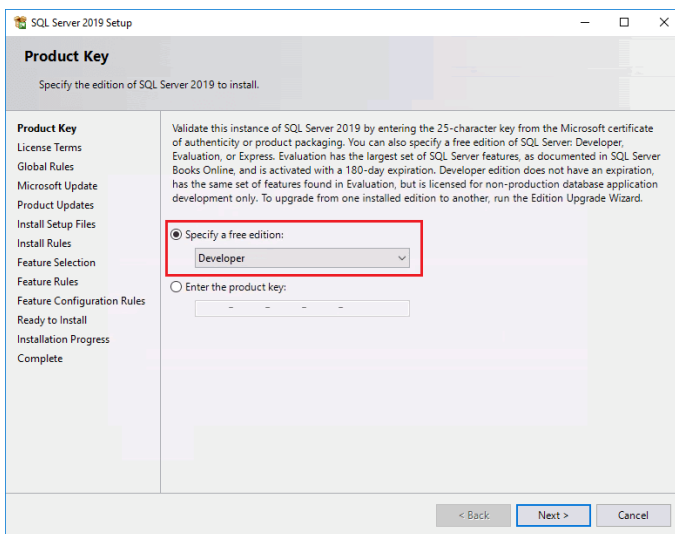
**Install SQL Server Management Tools**

Launch a download page that provides a link to install SQL Server Management Studio, SQL Server command-line utilities (SQLCMD and BCP), SQL Server PowerShell provider, SQL Server Profiler and Database Tuning Advisor. An internet connection is required to install these tools.

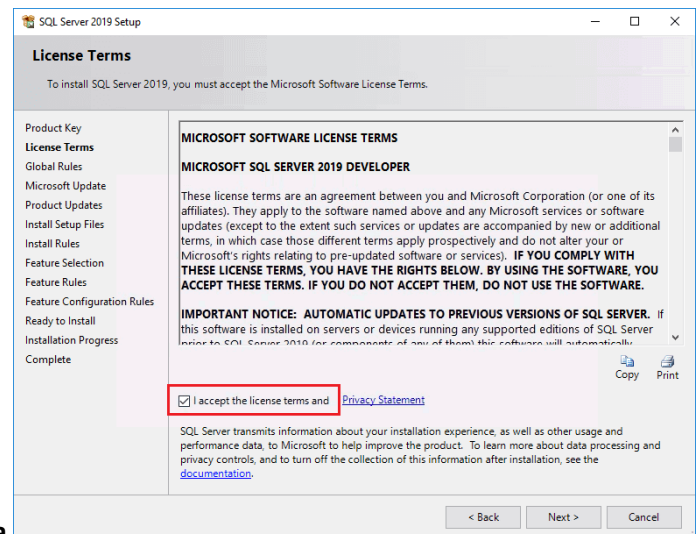
**Install SQL Server Data Tools**

Launch a download page that provides a link to install SQL Server Data Tools (SSDT). SSDT provides Visual Studio integration including project system support for Microsoft Azure SQL Database, the SQL Server Database Engine, Reporting Services, Analysis Services and Integration Services. An internet connection is required to install SSDT.

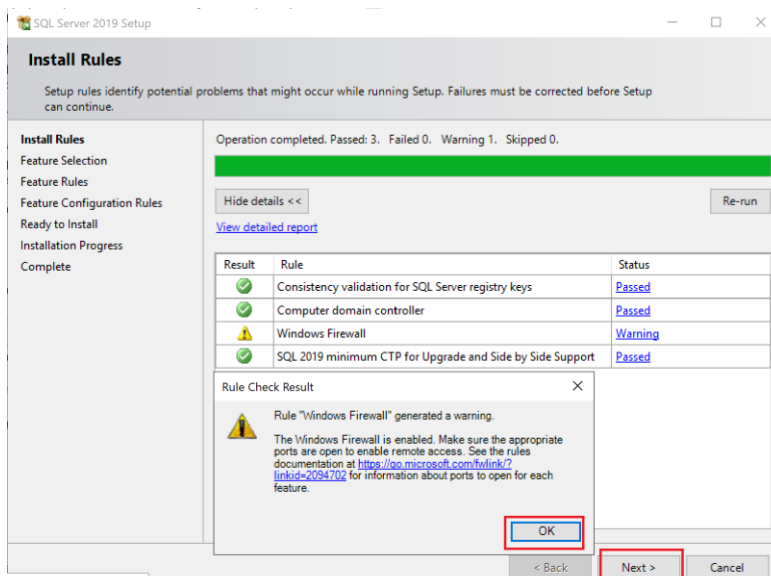
7.a



8.a



9.a



10.a

Features:

Instance Features

- ☒ Database Engine Services
 - ☐ SQL Server Replication
 - ☐ Machine Learning Services and Language Extensions
 - ☐ R
 - ☐ Python
 - ☐ Java
 - ☒ Full-Text and Semantic Extractions for Search
 - ☐ Data Quality Services
 - ☐ PolyBase Query Service for External Data
 - ☐ Java connector for HDFS data sources
- ☐ Analysis Services

Shared Features

- ☐ Machine Learning Server (Standalone)
 - ☐ R
 - ☐ Python
- ☐ Data Quality Client
- ☒ Client Tools Connectivity
- ☐ Integration Services
 - ☐ Scale Out Master
 - ☐ Scale Out Worker
- ☒ Client Tools Backwards Compatibility
- ☒ Client Tools SDK
- ☐ Distributed Replay Controller
- ☐ Distributed Replay Client
- ☒ SQL Client Connectivity SDK
- ☐ Master Data Services

Redistributable Features

11.a

12.a

SQL Server 2019 Setup

Instance Configuration

Specify the name and instance ID for the instance of SQL Server. Instance ID becomes part of the installation path.

Install Rules
Feature Selection
Feature Rules
Instance Configuration
Server Configuration
Database Engine Configuration
Feature Configuration Rules
Ready to Install
Installation Progress
Complete

☐ Default instance
☒ Named instance:

Instance ID:

SQL Server directory: C:\Program Files\Microsoft SQL Server\MSSQL15.MSSQLSERVER

Installed instances:

Instance Name	Instance ID	Features	Edition	Version

13.a

SQL Server 2019 Setup

Server Configuration

Specify the service accounts and collation configuration.

Install Rules
Feature Selection
Feature Rules
Instance Configuration
Server Configuration
Database Engine Configuration
Feature Configuration Rules
Ready to Install
Installation Progress
Complete

Service Accounts Collation

Microsoft recommends that you use a separate account for each SQL Server service.

Service	Account Name	Password	Startup Type
SQL Server Agent	NT Service\SQLAgent\$...		Manual
SQL Server Database Engine	NT Service\MSSQLSMIFI...		Automatic
SQL Full-text Filter Daemon Launc...	NT Service\MSSQLFDLa...		Manual
SQL Server Browser	NT AUTHORITY\LOCAL ...		Disabled

☐ Grant Perform Volume Maintenance Task privilege to SQL Server Database Engine Service

This privilege enables instant file initialization by avoiding zeroing of data pages. This may lead to information disclosure by allowing deleted content to be accessed.

[Click here for details](#)

< Back **Next >** Cancel

14.a

SQL Server 2019 Setup

Database Engine Configuration

Specify Database Engine authentication security mode, administrators, data directories, TempDB, Max degree of parallelism, Memory limits, and Filestream settings.

Install Rules
Feature Selection
Feature Rules
Instance Configuration
Server Configuration
Database Engine Configuration
Feature Configuration Rules
Ready to Install
Installation Progress
Complete

Server Configuration Data Directories TempDB MaxDOP Memory FILESTREAM

Specify the authentication mode and administrators for the Database Engine.

Authentication Mode

☐ Windows authentication mode
☒ Mixed Mode (SQL Server authentication and Windows authentication)

Specify the password for the SQL Server system administrator (sa) account.

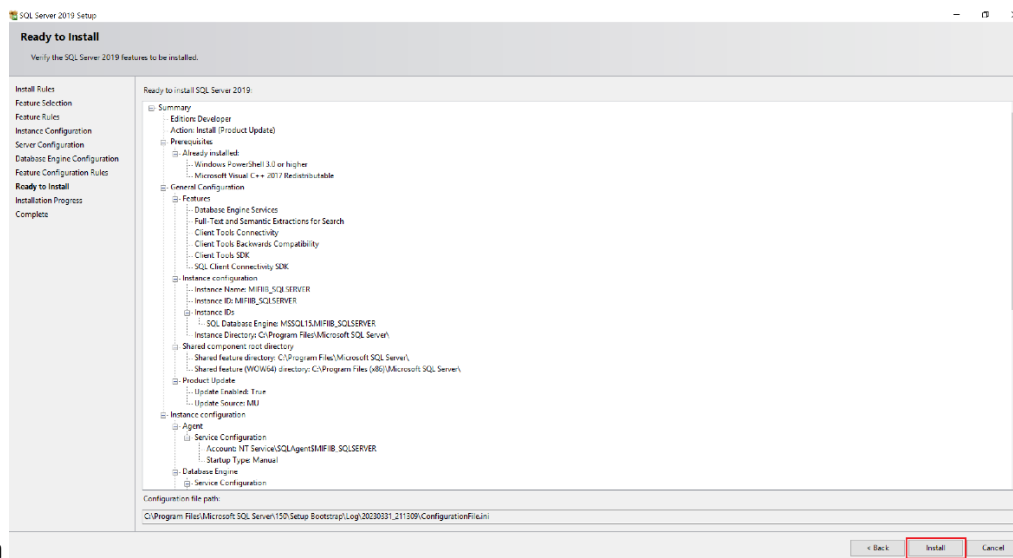
Enter password:
 Confirm password:

Specify SQL Server administrators

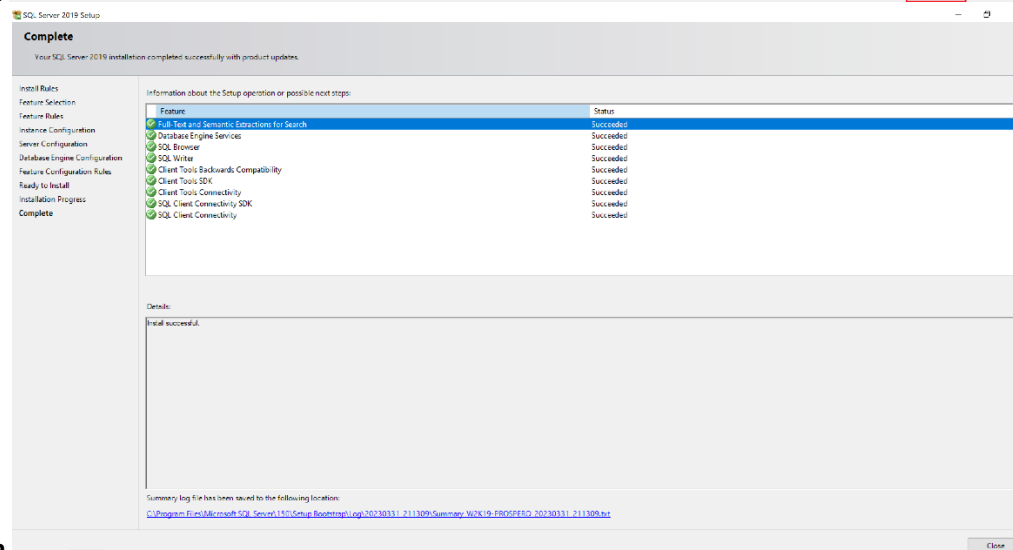
W2K19-PROSPERO\Administrator (Administrator)	SQL Server administrators have unrestricted access to the Database Engine.

Add Current User Add... Remove

< Back **Next >** Cancel

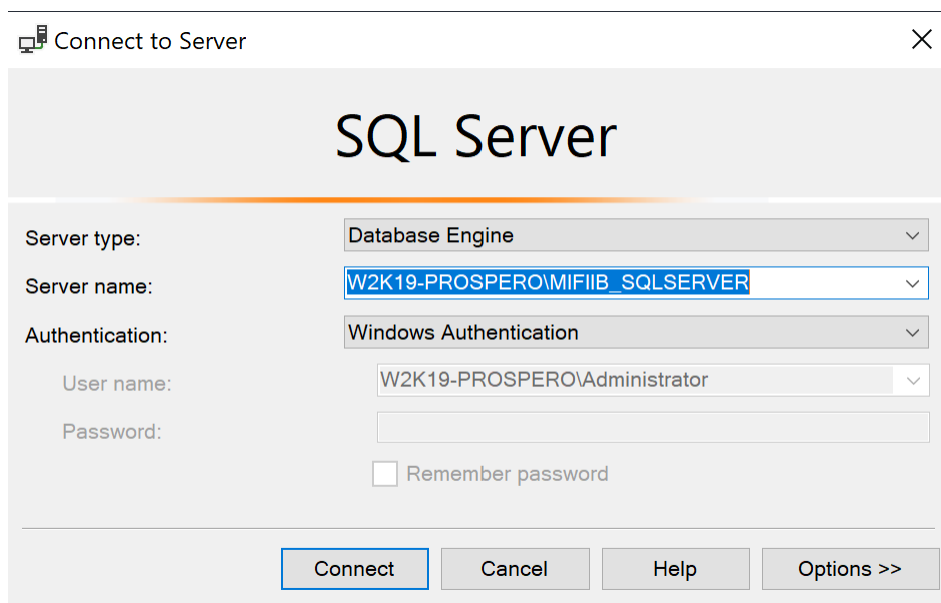


15.a

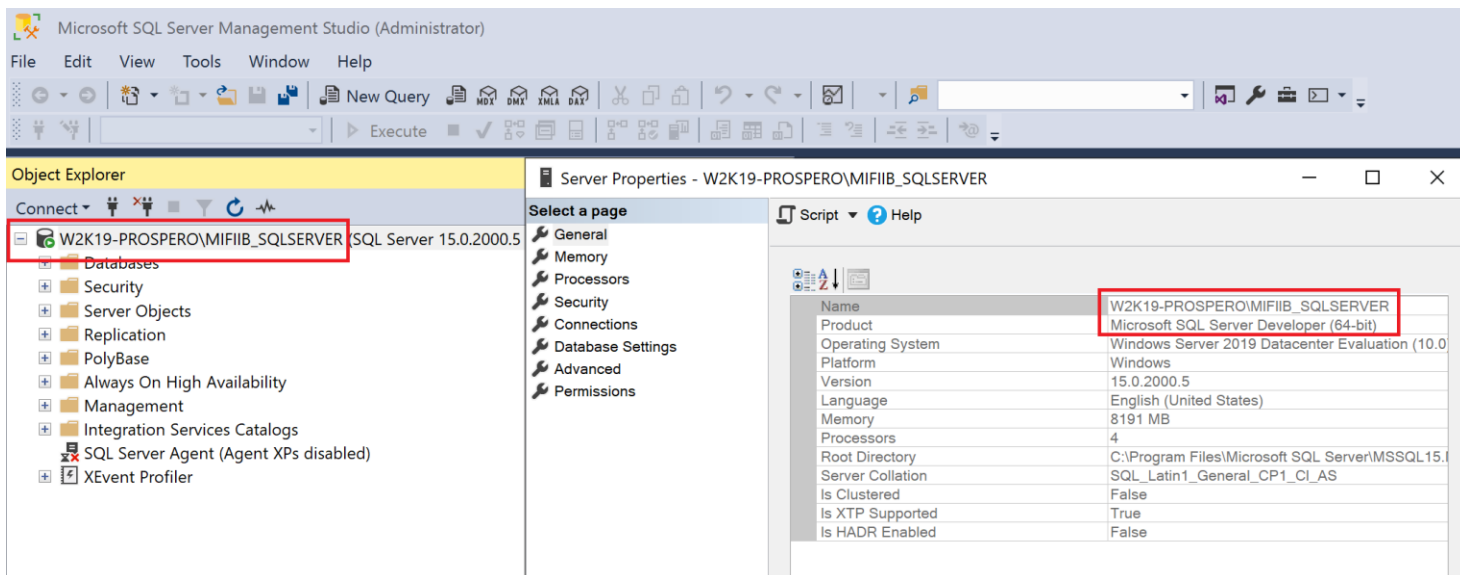


16.a

Установка SQL Server 2019 Developer Edition завершена, с помощью MSSMS (Microsoft SQL Server Management Studio) мы можем подключиться к нашему серверу, для управления (Pic#16):



Pic#16. Connect to Server

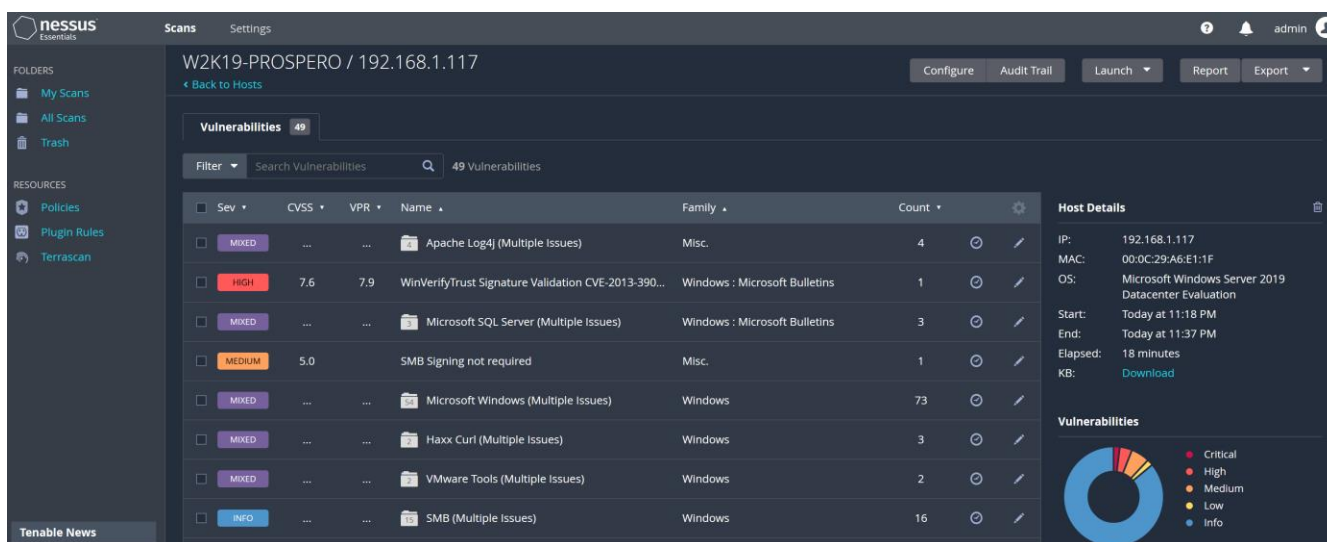


Pic#17. MS SQL Server 2019 Developer Edition

6. Базовое сканирование системы на наличие уязвимостей.

По условию ДЗ, мы должны установить MBSA (Microsoft Baseline Security Analyzer) и с помощью этой утилиты проверить операционную систему на наличие уязвимостей. С 2015 года компания Microsoft прекратила дальнейшую разработку данной утилиты. Операционные системы нового поколения (Windows Server 2K19/2K22, Windows 11) официально не поддерживаются этой утилитой.

По согласованию с ментором, допускается использование других инструментов для сканирования и генерации отчета, поэтому сканирование системы было проведено с помощью NextGen сканера уязвимостей **Nessus (Pic#18)**:



Pic#18. Host Security Report

Детальный отчет об обнаруженных уязвимостях прилагается, файл в формате html: **W2K19-PROSPERO_security_report.html**

Практическая работа завершена!